

Positivity 1.8.A

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1 Some commutative algebra

Definition 1. For a sequence $\underline{a} = (a_1, \dots, a_m)$ of elements of a commutative unitary ring A , the Koszul complex $K_\bullet(\underline{a})$ of A w.r.t. $\underline{a}$ is defined as

$$\cdots \longrightarrow K_2(\underline{a}) \longrightarrow K_1(\underline{a}) \longrightarrow K_0(\underline{a}) \longrightarrow 0$$

where $L = \bigoplus_{1 \leq i \leq m} Ae_i$ and $K_i(\underline{a}) = \wedge^i L$, where the boundary map is defined by

$$\begin{aligned} d_i : K_i(\underline{a}) &\longrightarrow K_{i-1}(\underline{a}) \\ e_{j_1} \wedge \cdots \wedge e_{j_i} &\longmapsto \sum_{k=1}^i (-1)^k a_{i_k} e_{j_1} \wedge \cdots \wedge \hat{e}_{j_k} \wedge \cdots \wedge e_{j_i} \end{aligned}$$

Given a A -module M , we call $K_\bullet(\underline{a}, M) := K_\bullet(\underline{a}) \otimes M$ the Koszul complex of M w.r.t. $\underline{a}$.

Proposition 1.1. Let $a_1 \in A^*$ be a unit, then $H_i(K_\bullet(\underline{a}, M)) = 0$ for all $i > 0$.

Proof. We define a homotopy map

$$\begin{aligned} h_i : K_i(\underline{a}, M) &\longrightarrow K_{i+1}(\underline{a}, M) \\ e_{j_1} \wedge \cdots \wedge e_{j_i} &\longmapsto a_1^{-1} e_1 \wedge e_{j_1} \wedge \cdots \wedge e_{j_i} \end{aligned}$$

Then one can easily check that

$$\text{id} = hd + dh$$

□

Corollary 1.1. For any A -module M , we have $H_i(K_\bullet(\underline{a}, M))_{a_j} = 0$ for all $i \geq 1$ and $1 \leq j \leq m$.

Proof. By 1.1 one can get $H_i(K_\bullet(\underline{a}, M_{a_j})) = 0$, and use the fact that localization functor is exact we have

$$H_i(K_\bullet(\underline{a}, M))_{a_j} = H_i(K_\bullet(\underline{a}, M_{a_j})) = 0$$

□

Definition 2. Let A be a commutative unitary ring and M be an A -module.

1. we say that $a \in A$ is M -regular if

$$M \xrightarrow{\cdot a} M$$

is injective.

2. we say that a sequence $(a_1, \dots, a_m) \in A^m$ is an M -regular sequence if

- a_1 is M -regular, a_2 is $\frac{M}{a_1 M}$ -regular, ..., and a_m is $\frac{M}{(a_1, \dots, a_{m-1})M}$ -regular.
- $M/(a_1, \dots, a_m)M \neq 0$.

Remark 1. Note that this definition corresponds to the usual definition in local ring $(A, \mathfrak{m})$ since the second condition means that $a_i \in \mathfrak{m}$.

Theorem 1.1. Let $\underline{a} = (a_1, \dots, a_m)$ be a sequence in a commutative unitary ring A and M be an A -module.

1. If $\underline{a}$ is an M -regular sequence, then $H_i(K_\bullet(\underline{a}, M)) = 0$ for all $i > 0$.
2. Conversely, assume that $(A, \mathfrak{m})$ is a Noetherian local ring, $a_i \in \mathfrak{m}$ and M is a finitely generated A -module. If $H_1(K_\bullet(\underline{a}, M)) = 0$, then $\underline{a}$ is an M -regular sequence.

Proof.

1. Induction on m , when $m = 1$ this is a direct result from the definition. For $m > 1$, we consider the exact sequence of complex

$$0 \longrightarrow K_\bullet(\underline{a}, M) \xrightarrow{\cdot a_1} K_\bullet(\underline{a}, M) \longrightarrow K_\bullet(\underline{a}', M_1) \oplus K_\bullet(\underline{a}', M_1)[-1] \longrightarrow \bullet$$

$\cdot a_1$ is injective since a_1 is M -regular, and

$$K_\bullet(\underline{a}', M_1) \oplus K_\bullet(\underline{a}', M_1)[-1] \cong \text{coker}(\cdot a_1)$$

by direct calculation. Because

$$H_{i+1}(K_\bullet(\underline{a}', M_1) \oplus K_\bullet(\underline{a}', M_1)[-1]) = H_{i+1}(K_\bullet(\underline{a}', M_1)) \oplus H_i(K_\bullet(\underline{a}', M_1)) = 0$$

by induction hypothesis and taking homology of above exact sequence one can deduce that $H_i(K_\bullet(\underline{a}, M)) = 0$ for all $i > 0$.

2. Induction on m , $m = 1$ is just the definition of Koszul complex. For $m > 1$, the condition $H_1(K_\bullet(\underline{a}, M)) = 0$ implies that if $x_1, \dots, x_n \in M$ such that $\sum_{i=1}^m a_i x_i = 0$, then there exists $y_{ij} \in M$ such that

$$x_p = \sum_{i=1}^{p-1} a_i y_{ip} - \sum_{j=p+1}^m a_j y_{pj}$$

First consider $p = m$, we find that a_m is $\frac{M}{(a_1, \dots, a_{m-1})M}$ -regular. Now we proof that $\underline{a}' := (a_1, \dots, a_{m-1})$ is M -regular sequence. We claim that

$$H_1(K_\bullet(\underline{a}', M)) = \mathfrak{m}H_1(K_\bullet(\underline{a}', M))$$

then by Nakayama lemma one has $H_1(K_\bullet(\underline{a}', M)) = 0$, and the result from the induction hypothesis. For this, we proof that

$$H_1(K_\bullet(\underline{a}', M)) \subseteq a_m \cdot H_1(K_\bullet(\underline{a}', M)) \subseteq \mathfrak{m}H_1(K_\bullet(\underline{a}', M))$$

Pick any $(x_1, \dots, x_{m-1}) \in K_1(\underline{a}', M)$ such that $\sum_{i=1}^{m-1} a_i x_i = 0$, then by above formula about x_p (here we take $x_m = 0$) we find that

$$x_p + a_m y_{pm} = \sum_{i=1}^{p-1} a_i y_{ip} - \sum_{j=p+1}^{m-1} a_j y_{pj}$$

This gives the desired result.

□

Corollary 1.2. Let $(A, \mathfrak{m})$ be a commutative unitary Noetherian local ring, $a_1, \dots, a_m \in \mathfrak{m}$ and M be a finitely generated A -module. Then we have

1. $\underline{a} = (a_1, \dots, a_m)$ is M -regular sequence iff $H_1(K_\bullet(\underline{a}, M)) = 0$.
2. If $(a_1, \dots, a_m)$ is an M -regular sequence, then for any $\sigma \in S_m$, $(a_{\sigma(1)}, \dots, a_{\sigma(m)})$ is also an M -regular sequence.

Now we switch to the Koszul complex of locally free sheaves.

Definition 3. For a locally free sheaf $\mathcal{E}$ of rank m over a scheme X and a global section $s \in H^0(X, \mathcal{E})$ (or equivalently, a morphism $s : \mathcal{E}^\vee \rightarrow \mathcal{O}_X$), the corresponding Koszul complex is

$$\cdots \longrightarrow \wedge^2 \mathcal{E}^\vee \longrightarrow \wedge^1 \mathcal{E}^\vee \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{Z(s)} \longrightarrow 0$$

where $Z(s)$ is the zero scheme of s , defined by ideal sheaf $\mathcal{I}_{Z(s)} = \text{im}(s^\vee : \mathcal{E}^\vee \rightarrow \mathcal{O}_X)$. i.e. $x \in Z(s)$ iff $s(x) = 0$ in $\mathcal{E} \otimes k(x)$. Here the boundary morphism is given by

$$\begin{aligned} d_i : \wedge^{i+1} \mathcal{E}^\vee &\longrightarrow \wedge^i \mathcal{E}^\vee \\ e_{j_1} \wedge \cdots \wedge e_{j_i} &\longmapsto \sum_{k=1}^i (-1)^k s(e_{j_k}) e_{j_1} \wedge \cdots \wedge \hat{e}_{j_k} \wedge \cdots \wedge e_{j_i} \end{aligned}$$

Proposition 1.2. The above sequence is exact at $X - Z(s)$. In particular, if s is surjectivem, then the Koszul complex is exact.

Proof. For any $x \in X - Z(s)$, we localize the above complex at x and note that it just the Koszul complex w.r.t. the sequence $(s_x(e_1), \dots, s_x(e_m))$ for a basis $\{e_i\}$ of $\mathcal{E}_x^\vee$ and $s_x : \mathcal{E}_x^\vee \rightarrow \mathcal{O}_{X,x}$. And the condition $x \notin Z(s)$ means that there exists $s_x(e_i) \notin \mathfrak{m}_{X,x}$ i.e. $s_x(e_i)$ is a unit of $\mathcal{O}_{X,x}$. Now use 1.1. □

Proposition 1.3. If X is locally Cohen-Macaulay scheme and $\text{codim}_X Z = \text{rk } \mathcal{E}$, then the Koszul complex described above is exact.

Proof. we just localize above Koszul complex at x and note that for a CM ring $\mathcal{O}_{X,x}$, a sequence $(a_1, \dots, a_m)$ is a regular sequence iff $\dim \mathcal{O}_{X,x}/(a_1, \dots, a_m) = \dim \mathcal{O}_{X,x} - m$, which is ensured by $\text{codim}_X Z = \text{rk } \mathcal{E}$. And finally use 1.2. □

We give a generalize version of the Koszul complex as follows.

Definition 4. (Eagon-Northcott complex) Suppose we have a morphism of locally free sheaves $\varphi : \mathcal{E} \rightarrow \mathcal{F}$ on a scheme X , where $\text{rk } \mathcal{E} = e$, $\text{rk } \mathcal{F} = f$ and $e \geq f$, then we have the complex $\text{EN}(\varphi)_k$, which resolves $S^k \mathcal{F}$:

$$0 \longrightarrow L_p \longrightarrow L_{p-1} \longrightarrow \cdots \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0$$

where

$$L_p = \begin{cases} (\wedge^p \mathcal{E}) \otimes (S^{k-p} \mathcal{F}), & 0 \leq p \leq k \\ (\wedge^{f+p-1} \mathcal{E}) \otimes (S^{p-k-1} \mathcal{F}^*) \otimes (\wedge^f \mathcal{F}^*), & p \geq k+1 \end{cases}$$

Proposition 1.4. Assumption as above, and suppose more that the ideal $\mathcal{J}(\varphi)$ is generated by all determinants of all possible $f \times f$ sub-matrices of φ and $Z(\varphi)$ is the closed subscheme determined by $\mathcal{J}(\varphi)$. then we have $\text{EN}(\varphi)_k$ is exact at $X - Z(\varphi)$ for all $k \geq 0$. In particular, if φ is surjective, then $\text{EN}(\varphi)_k$ is exact.

Proof. omit. □

Finally we introduce some proposition of finitely generated graded modules and its free resolutions. Here the rings are $\mathbb{N}$ -graded and modules are $\mathbb{Z}$ -graded.

Definition 5. Let S be a graded ring, a resolution is call a graded free resolution if it is exact, all terms are graded free modules and the maps are all homogeneous maps of degree 0.

For the rest part, we assume that $S_0 = k$ is a field.

Definition 6. A complex of graded modules

$$\cdots \longrightarrow E_n \xrightarrow{\varphi_n} E_{n-1} \longrightarrow \cdots$$

over a graded ring S with $S_0 = k$ is called minimal complex if $\varphi_n(E_n) \subseteq S_+ E_{n-1}$. i.e. we do not allow constant generators.

Remark 2. A minimal complex need not to be exact

$$0 \longrightarrow S \xrightarrow{0} S \longrightarrow 0$$

Definition 7. A complex is a trivial complex if it is the direct sum of complex of the form

$$\cdots \longrightarrow 0 \longrightarrow S(-d) \xrightarrow{c \cdot \text{id}_{S(-d)}} S(-d) \longrightarrow 0 \longrightarrow \cdots$$

here $c \in S_0 = k$.

Theorem 1.2. (Existence of minimal free resolution) Let S be a graded ring with $S_0 = k$, and

$$\cdots \longrightarrow E_n \xrightarrow{\varphi_n} E_{n-1} \longrightarrow \cdots \longrightarrow E_0$$

is a graded free resolution over S . Then it is a minimal iff for each n , a basis of E_{n-1} maps onto a minimal set of generators of $\text{coker } \varphi_n$. In particular, for any finitely generated graded module, the minimal resolution exists by the graded Nakayama lemma.

Proof. Omit. See GTM150 lemma 19.4. □

Theorem 1.3. (Uniqueness of minimal free resolution) Let S be a graded ring with $S_0 = k$, and F be a finitely generated graded S -module. If $\mathcal{C}_\bullet$ is a minimal graded free resolution of F , then any graded free resolution of F is isomorphic to the direct sum of $\mathcal{C}_\bullet$ and a trivial complex. In particular, the uniqueness says that if

$$\cdots \longrightarrow E_n \longrightarrow \cdots \longrightarrow E_2 \longrightarrow E_1 \longrightarrow E_0 \longrightarrow F \longrightarrow 0$$

is a minimal free resolution with $E_p = \oplus S(-a_{p,i})$ given by matrix of homogeneous polynomial containing no non-zero constants as entries e.g. constructed by Nakayama lemma, then $a_{p,i}$ are uniquely determined by F .

Proof. Omit. See GTM150 theorem 20.2. □

Theorem 1.4. (Hilbert syzygy theorem) Let k be a field, then any finitely generated graded module over $k[x_1, \dots, x_n]$ has a graded free resolution of length $\leq n$.

Proof. Omit. See GTM150 corollary 19.7. □

2 Definition and property of CM regularity

Definition 8. (Regularity w.r.t. a map) Let $f : X \rightarrow Y$ be a proper surjective morphism of schemes and $\mathcal{L}$ is a invertible sheaf on X such that

1. $\mathcal{L}$ is f -ample;
2. $f^*f_*\mathcal{L} \rightarrow \mathcal{L}$ is surjective.

For a coherent sheaf $\mathcal{F} \in \text{Coh}(X)$, we say that $\mathcal{F}$ is m -regular with respect to $\mathcal{L}$ and f if

$$R^i f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m-i)}) = 0$$

for all $i > 0$. We denote $\text{reg}(\mathcal{F}) = \min\{m \mid \mathcal{F} \text{ is } m\text{-regular}\}$ and if $\dim \mathcal{F} = 0$ then we denote $\text{reg}(\mathcal{F}) = -\infty$.

Theorem 2.1. (Mumford's theorem) Let $X, Y, f, \mathcal{L}, \mathcal{F}$ be as above, then we have

1. The morphism

$$f^*f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m+k)}) \rightarrow \mathcal{F} \otimes \mathcal{L}^{\otimes m+k}$$

is surjective;

2. The morphism

$$f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes f_*(\mathcal{L}^{\otimes k}) \rightarrow f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m+k)})$$

is surjective;

3. $\mathcal{F}$ is $(m+k)$ -regular with respect to $\mathcal{L}$ and f

Proof. Note that these property are all affine local on Y (surjectivity and higher direct image). thus we assume that $Y = \text{Spec}(A)$ is affine.

1. By the relative Cartan-Serre-Grothendieck theorem (Lazarsfeld theorem 1.7.6) we know that

$$f^*f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes n}) \rightarrow \mathcal{F} \otimes \mathcal{L}^{\otimes n}$$

is surjective for all $n \geq n_0$ (we may assume $n_0 \geq m+k$). By the third part of Mumford theorem we know $\mathcal{F}$ is $m+k$ -regular, and thus by the second part of Mumford's theorem and f^* commute with tensor product we know that

$$f^*f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m+k)}) \otimes f^*f_*(\mathcal{L}^{\otimes(n_0-m-k)}) \rightarrow f^*f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes n_0}) \rightarrow \mathcal{F} \otimes \mathcal{L}^{\otimes n_0} = (\mathcal{F} \otimes \mathcal{L}^{\otimes m+k}) \otimes (\mathcal{L}^{\otimes n_0-m-k}).$$

Now part one is derived by the lemma below 2.1.

2. We use the third part to reduce to the case $k = 1$, since the below diagram.

$$\begin{array}{ccc} f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes f_*(\mathcal{L}^{\otimes k}) \otimes f_*(\mathcal{L}) & \longrightarrow & f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m+k}) \otimes f_*(\mathcal{L}) \\ \downarrow & & \downarrow \\ f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes f_*(\mathcal{L}^{\otimes k+1}) & \longrightarrow & f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m+k+1}) \end{array}$$

Now since f is proper, we know that $f_*\mathcal{F} = \tilde{M}$ is a coherent sheaf. Thus one can choose a finite set of generators $f_1, \dots, f_r$ of M as A -module. Let $V = \text{span}_k\{f_1, \dots, f_r\}$, now we have a surjective

$$V \otimes_k A \twoheadrightarrow M$$

take $\sim$ and pullback we get a surjective morphism

$$V \otimes \mathcal{O}_X \twoheadrightarrow f^*f_*\mathcal{L} \twoheadrightarrow \mathcal{L}$$

We consider the Koszul exact sequence corresponding to $V \otimes \mathcal{O}_X \otimes \mathcal{L}^\vee \twoheadrightarrow \mathcal{O}_X$,

$$\begin{array}{ccccccc} \dots & \longrightarrow & \wedge^2(V \otimes \mathcal{O}_X \otimes \mathcal{L}^{\otimes(-1)}) & \longrightarrow & \wedge^1(V \otimes \mathcal{O}_X \otimes \mathcal{L}^{\otimes(-1)}) & \longrightarrow & \mathcal{O}_X \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ \dots & \longrightarrow & (\wedge^2 V) \otimes \mathcal{L}^{\otimes(-2)} & \longrightarrow & (\wedge^1 V) \otimes \mathcal{L}^{\otimes(-1)} & \longrightarrow & \mathcal{O}_X \longrightarrow 0 \end{array}$$

Tensoring the above exact sequence by $\mathcal{F} \otimes \mathcal{L}^{\otimes(m+1)}$

$$\begin{array}{ccccccc} & & 0 & & & & 0 \\ & & \searrow & & \nearrow & & \\ & & & \text{im } d_2 & & & \\ & & \nearrow & & \searrow & & \\ \dots & \longrightarrow & (\wedge^2 V) \otimes \mathcal{F} \otimes \mathcal{L}^{\otimes(m-1)} & \longrightarrow & (\wedge^1 V) \otimes \mathcal{F} \otimes \mathcal{L}^{\otimes m} & \longrightarrow & \mathcal{F} \otimes \mathcal{L}^{\otimes(m+1)} \longrightarrow \dots \\ & \searrow & \nearrow & & & & \\ & & \text{im } d_3 & & & & \\ & \nearrow & & & & & \\ 0 & & & & & & \end{array}$$

Consider the first short exact sequence we have that the below complex is exact.

$$f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes V = f_*(V \otimes \mathcal{F} \otimes \mathcal{L}^{\otimes m}) \rightarrow f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m+1)}) \rightarrow R^1 f_* \text{im } d_2$$

Consider the i -th short exact sequence we have

$$R^1 f_* \text{im } d_2 \hookrightarrow R^2 f_* \text{im } d_3 \hookrightarrow \dots \hookrightarrow R^i f_* \text{im } d_{i+1} = 0$$

when $i > \dim X$. Thus

$$f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes V \twoheadrightarrow f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m+1)})$$

but we know this morphism is given by applying f_* to

$$(V \otimes \mathcal{O}_X) \otimes (\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \rightarrow f^*f_*\mathcal{L} \otimes (\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \rightarrow \mathcal{L} \otimes (\mathcal{F} \otimes \mathcal{L}^{\otimes m})$$

thus

$$f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes V \longrightarrow f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes f_*(\mathcal{L}) \twoheadrightarrow f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m+1)})$$

and the conclusion is done.

3. We use the third part to reduce to the case $k = 1$. Tensoring with $\mathcal{F} \otimes \mathcal{L}^{\otimes m}$ will get

$$\begin{array}{ccccccc}
 & & 0 & & & & 0 \\
 & & \searrow & & \nearrow & & \\
 & & & \text{im } d_2 & & & \\
 & & \nearrow & & \searrow & & \\
 \dots & \longrightarrow & (\wedge^2 V) \otimes \mathcal{F} \otimes \mathcal{L}^{\otimes(m-2)} & \longrightarrow & (\wedge^1 V) \otimes \mathcal{F} \otimes \mathcal{L}^{\otimes(m-1)} & \longrightarrow & \mathcal{F} \otimes \mathcal{L}^{\otimes m} \longrightarrow \dots \\
 & \searrow & \nearrow & & & & \\
 & & \text{im } d_3 & & & & \\
 0 & \nearrow & & & & &
 \end{array}$$

Similarly, by splitting into short exact sequence and take higher direct image, we will have that

$$R^i f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m-i)}) \hookrightarrow R^{i+1} f_*(\text{im } d_1) \hookrightarrow \dots \hookrightarrow R^{i+k} f_*(\text{im } d_k) = 0$$

when $i + k > \dim X$.

□

Lemma 2.1. If $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ are sheaves of $\mathcal{O}_X$ -module, with morphism

$$f : \mathcal{A} \rightarrow \mathcal{C}, \quad g : \mathcal{B} \rightarrow \mathcal{D}$$

such that

1. the induce map $f \otimes g : \mathcal{A} \otimes \mathcal{B} \rightarrow \mathcal{C} \otimes \mathcal{D}$ is surjective.
2. $\mathcal{D}$ is an invertible sheaf.
3. g is surjective.

Then f is surjective.

Proof. Consider the following diagram

$$\begin{array}{ccccc}
 & & f \otimes g & & \\
 & \nearrow & & \searrow & \\
 \mathcal{A} \otimes \mathcal{B} & \xrightarrow{\text{id} \otimes g} & \mathcal{A} \otimes \mathcal{D} & \xrightarrow{f \otimes \text{id}} & \mathcal{C} \otimes \mathcal{D}
 \end{array}$$

we find that $f \otimes \text{id}$ is surjective. Now tensoring with $\text{id}^\vee$ we know that f is surjective since

$$\mathcal{D} \otimes \mathcal{D}^\vee \cong \mathcal{O}_X$$

□

Example 2.1. Take $Y = \text{Spec}(k)$ then we get that $\mathcal{F}$ is m -regular iff

$$H^i(X, \mathcal{F} \otimes \mathcal{L}^{\otimes(m-i)}) = 0$$

for all $i > 0$, since $R^i f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m-i)}) = H^i(X, \mathcal{F})^\sim$ if Y is affine.

If $\mathcal{F}$ is m -regular, then we have

1. $\mathcal{F} \otimes \mathcal{L}^{\otimes(m+k)}$ is generated by global sections.

2. The natural maps

$$H^0(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes H^0(X, \mathcal{L}^{\otimes k}) \rightarrow H^0(X, \mathcal{F} \otimes \mathcal{L}^{\otimes(m+k)})$$

are surjective.

3. $\mathcal{F}$ is $(m+k)$ -regular.

We list some proposition.

Proposition 2.1. If $\mathcal{F}_i$ is m -regular with respect to $\mathcal{L}$ and f , then $\varinjlim \mathcal{F}_i$ is m -regular with respect to $\mathcal{L}$ and f . This is ensured by the fact that for a morphism $f : X \rightarrow Y$ of Noetherian schemes (or more generally f is qcqs), we have

$$R^i f_* \varinjlim \mathcal{G}_i \cong \varinjlim R^i f_* \mathcal{G}_i$$

And conversely, if finite direct limit $\varinjlim \mathcal{F}_i$ is m -regular, then $\mathcal{F}_i$ are all m -regular.

Proposition 2.2. Suppose that a coherent sheaf $\mathcal{F}$ on X is resolved by a long exact sequence

$$\cdots \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F} \longrightarrow 0$$

of coherent sheaves on X . If $\mathcal{F}_i$ is $(m+i)$ -regular with respect to $\mathcal{L}$ and f for every $i \geq 0$, then $\mathcal{F}$ is m -regular with respect to $\mathcal{L}$ and f .

Proof. We decompose the exact sequence into short exact sequence.

$$\begin{array}{ccccccc}
 & & 0 & & & & \\
 & & \searrow & & & & \\
 & & \text{im } d_2 & & & & \\
 & \nearrow & \searrow & & & & \\
 \cdots & \longrightarrow & \mathcal{F}_2 & \longrightarrow & \mathcal{F}_1 & \longrightarrow & \mathcal{F}_0 \longrightarrow \mathcal{F} \longrightarrow 0 \\
 & & & & \searrow & \nearrow & \\
 & & & & \text{im } d_1 & & \\
 & & & & \nearrow & \searrow & \\
 & & 0 & & & & 0
 \end{array}$$

Twist each short exact sequence by $\mathcal{L}^{\otimes(m-i)}$ and take higher direct image to get

$$R^i f_*(\mathcal{F}_0 \otimes \mathcal{L}^{\otimes(m-i)}) \longrightarrow R^i f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m-i)}) \longrightarrow R^{i+1} f_*(\text{im } d_1 \otimes \mathcal{L}^{\otimes((m+1)-(i+1))}) \longrightarrow R^{i+1} f_*(\mathcal{F}_0 \otimes \mathcal{L}^{\otimes((m+1)-(i+1))})$$

since $\mathcal{F}_0$ is m -regular and by Mumford's theorem it's $m+1$ -regular, thus we have

$$R^i f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m-i)}) = R^{i+1} f_*(\text{im } d_1 \otimes \mathcal{L}^{\otimes((m+1)-(i+1))})$$

continue this process, we find that

$$R^i f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes(m-i)}) = R^{i+1} f_*(\text{im } d_1 \otimes \mathcal{L}^{\otimes((m+1)-(i+1))}) = \cdots = R^{i+k} f_*(\text{im } d_k \otimes \mathcal{L}^{\otimes((m+k)-(i+k))}) = 0$$

for $i+k > \dim X$. □

For the rest part, we only consider the case $Y = \text{Spec}(k)$ and for simplicity we denote $\mathcal{F} \otimes \mathcal{L}^{\otimes m}$ resp. $\mathcal{L}$ by $\mathcal{F}(m)$ resp. $\mathcal{O}_X(1)$.

Proposition 2.3. (Linear resolutions) Let $X = \mathbb{P}^n$ be a projective scheme with ample line bundle $\mathcal{L}$ and $\mathcal{F}$ be a coherent sheaf on X . Suppose more that X and $\mathcal{L}$ satisfies

$$H^i(X, \mathcal{O}_X) = H^i(X, \mathcal{L}^{\otimes(-i)}) = 0$$

for all $i, n > 0$, then $\mathcal{F}$ is m -regular with respect to f iff $\mathcal{F}$ is resolved by a long exact sequence

$$\cdots \longrightarrow \oplus \mathcal{O}_X(-m-2) \longrightarrow \oplus \mathcal{O}_X(-m-1) \longrightarrow \oplus \mathcal{O}_X(-m) \longrightarrow \mathcal{F} \longrightarrow 0$$

Proof. The if part is given by the above proposition. For the only if part, we set $W = H^0(\mathbb{P}, \mathcal{F}(m))$. Because $\mathcal{F}(m)$ is globally generated by Mumford theorem, we have a surjective morphism $W \otimes \mathcal{O}_X \twoheadrightarrow \mathcal{F}(m)$ i.e. $W \otimes \mathcal{O}_X(-m) \twoheadrightarrow \mathcal{F}$. Let $\mathcal{F}_1$ be the kernel of $\mathcal{O}_X(-m) \twoheadrightarrow \mathcal{F}$ i.e.

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow W \otimes \mathcal{O}_X(-m) \longrightarrow \mathcal{F} \longrightarrow 0$$

We just need to proof that $\mathcal{F}_1$ is $(m+1)$ -regular, since if it holds, we can continue the process to construct our desired exact sequence. For $i = 1$, we know that

$$H^0(X, W \otimes \mathcal{O}_X) = W \otimes H^0(X, \mathcal{O}_X) = W \xrightarrow{=} H^0(X, \mathcal{F}(m))$$

is surjective. Thus $H^1(X, \mathcal{F}_1((m+1)-1)) = 0$ by twisting $\mathcal{O}_X(m)$ to above short exact sequence and taking cohomology. For $i > 1$, twisting $\mathcal{O}_X(m-i+1)$ to above short exact sequence and taking cohomology we find

$$\begin{array}{c} 0 = W \otimes H^{i-1}(X, \mathcal{O}_X(1-i)) \rightarrow H^{i-1}(X, \mathcal{F}(m-(i-1))) \longrightarrow \\ \delta \\ \longleftarrow H^i(X, \mathcal{F}_1(m+1-i)) \longrightarrow W \otimes H^i(X, \mathcal{O}_X(1-i)) = 0 \end{array}$$

And thus

$$H^{i+1}(X, \mathcal{F}_1((m+1)-i)) = H^i(X, \mathcal{F}(m-i)) = 0.$$

□

Proposition 2.4. (Regularity of tensor products) Let $\mathcal{F}$ be a coherent sheaf on X , and $\mathcal{E}$ be a locally free sheaf on X . If $\mathcal{F}$ is m -regular and $\mathcal{E}$ is l -regular, then $\mathcal{F} \otimes \mathcal{E}$ is $(l+m)$ -regular. Moreover, we have

$$H^0(X, \mathcal{F}(m)) \otimes H^0(X, \mathcal{E}(l)) \rightarrow H^0(X, \mathcal{F} \otimes \mathcal{E}(m+l))$$

is surjective.

Proof. The first part is trivial by above proposition since $\otimes_{\mathcal{O}_X} \mathcal{E}$ is exact by the locally freeness of $\mathcal{E}$. For the second part, we look at the short exact sequence given in the proof and apply $\otimes \mathcal{E}(l)$

$$0 \longrightarrow \mathcal{F}_1(m) \otimes \mathcal{E}(l) \longrightarrow W \otimes \mathcal{O}_X \otimes \mathcal{E}(l) \longrightarrow \mathcal{F}(m) \otimes \mathcal{E}(l) \longrightarrow 0$$

□

Take cohomology we find that

$$H^0(X, \mathcal{F}(m)) \otimes H^0(X, \mathcal{E}(l)) \rightarrow H^0(X, \mathcal{F} \otimes \mathcal{E}(m+l)) \rightarrow H^1(X, \mathcal{F}_1 \otimes \mathcal{E}((l+m+1)-1)) = 0$$

Proposition 2.5. (Wedge and symmetric products) If $\mathcal{E}$ is an m -regular locally free sheaf, then $T^p\mathcal{E}$, $\wedge^p\mathcal{E}$ and $S^p\mathcal{E}$ are all (pm) -regular

Proof. This is a direct result of above propostion and the fact that in char $k = 0$, the $\wedge^p\mathcal{E}$ and $S^p\mathcal{E}$ are both direct sum of $T^p\mathcal{E}$ by the following lemma. $\square$

Lemma 2.2. Let k a field of character 0, and V a finite dimensional vector space, then we have that $\wedge^p V$ and $S^p V$ are both direct summand of $T^p V$.

Proof. We consider the following two linear maps

$$\begin{aligned} S_p : T^p V &\longrightarrow T^p V \\ v &\longmapsto \frac{1}{p!} \sum_{\sigma \in S_p} \sigma(v) \\ \wedge_p : T^p V &\longrightarrow T^p V \\ v &\longmapsto \frac{1}{p!} \sum_{\sigma \in S_p} (-1)^{\text{sgn}(\sigma)} \cdot \sigma(v) \end{aligned}$$

Note that for any $v \in S^p V$ resp. $w \in \wedge^p V$ we have

$$\sigma(v) = v \quad \text{resp.} \quad \sigma(w) = (-1)^{\text{sgn}(\sigma)} w$$

thus $\text{im } S_p = S^p V$ and $\text{im } \wedge_p = \wedge^p V$. Another direct calculation shows that $S_p^2 = S_p$ and $\wedge_p^2 = \wedge_p$. Thus

$$T^p V = \text{im } S_p \otimes \ker S_p = S^p V$$

$$T^p V = \text{im } \wedge_p \otimes \ker \wedge_p = \wedge^p V$$

$\square$

Now we give some examples

Example 2.2. For the simplest case $X = \mathbb{P}^n$, and $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(1)$, we have the following facts

1. The line bundle $\mathcal{O}_{\mathbb{P}^n}(a)$ is $(-a)$ -regular.
2. The ideal sheaf $\mathcal{I}_H$ of a linear subspace $H \subseteq \mathbb{P}^n$ is 1-regular.
3. If $i : H \rightarrow \mathbb{P}^n$ is a hypersurface of degree d , then $i_*\mathcal{O}_H$ is $(d-1)$ -regular.

Proof. 1. Trivial since we know all the cohomology group of $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(a))$. (See Hartshorne chapter 3 section 5).

2. If we denote $H = \bigcap_{i=1}^r V(s_i)$ with $s_i \in H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))$, then we can construct $(s_1, \dots, s_r) \in H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus r})$ and get a Koszual exact sequence 1.3, where $\mathcal{E} = \mathcal{O}_X(1)^{\oplus r}$

$$\begin{array}{ccccccccccccccc} 0 & \longrightarrow & \wedge^r \mathcal{E}^\vee & \longrightarrow & \wedge^{r-1} \mathcal{E}^\vee & \longrightarrow & \dots & \longrightarrow & \wedge^2 \mathcal{E}^\vee & \longrightarrow & \wedge^1 \mathcal{E}^\vee & \longrightarrow & \mathcal{O}_{\mathbb{P}^n} & \longrightarrow & \mathcal{O}_H & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{O}_X(-r) & \longrightarrow & \mathcal{O}_X(-r+1)^r & \longrightarrow & \dots & \longrightarrow & \mathcal{O}_X(-2)^{\binom{r+1}{2}} & \longrightarrow & \mathcal{O}_X(-1)^r & \longrightarrow & \mathcal{O}_{\mathbb{P}^n} & \longrightarrow & \mathcal{O}_H & \longrightarrow & 0 \\ & & & & & & & & & & \downarrow & & \nearrow & & & & \\ & & & & & & & & & & \mathcal{I}_H & & & & & & \\ & & & & & & & & & & \downarrow & & & & & & \\ & & & & & & & & & & 0 & & & & & & \end{array}$$

Now use the first part and 2.2.

3. We use the first part, the exact sequence of closed subscheme as follows and 2.2

$$0 \longrightarrow \mathcal{O}_X(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow i_*\mathcal{O}_H \longrightarrow 0$$

□

Theorem 2.2. (Green' theorem) Let $\mathbb{P} = \mathbb{P}(V)$ and $W \subseteq H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(d))$ be a subspace of codimension c giving a free linear series. Then the natural map

$$s_k : W \otimes H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(k)) \rightarrow H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(d+k))$$

is surjective for all $k \geq c$. i.e. the ideal generated by W in the ring $\Gamma(\mathcal{O}_{\mathbb{P}})$ contains all homogeneous polynomials of degree $\geq d+c$.

Proof. We have $S^k V = H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(k))$ and by above example $\mathcal{O}_{\mathbb{P}}(k)$ is $(-k)$ -regular, which implies that $\mathcal{O}_{\mathbb{P}}(k)$ is generated by global section. Let M_d be the kernel of the surjection $S^d V \otimes \mathcal{O}_{\mathbb{P}} \twoheadrightarrow \mathcal{O}_{\mathbb{P}}(d)$ and by taking cohomology of

$$0 \longrightarrow M_d \longrightarrow S^d V \otimes \mathcal{O}_{\mathbb{P}} \longrightarrow \mathcal{O}_{\mathbb{P}}(d) \longrightarrow 0$$

We know that M_d is 1-regular. Similarly we form the short exact sequence by the freeness of W (Recall Lazarsfeld definition 1.1.8 that the W is free means that $W \otimes_{\mathbb{C}} \mathcal{O}_{\mathbb{P}}(-d) \rightarrow \mathcal{O}_{\mathbb{P}}$ is surjective).

$$0 \longrightarrow M_W \longrightarrow W \otimes \mathcal{O}_{\mathbb{P}} \longrightarrow \mathcal{O}_{\mathbb{P}}(d) \longrightarrow 0$$

Now consider the following diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & \ker \varphi & \longrightarrow & 0 & \longrightarrow & 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & M_W & \longrightarrow & W \otimes \mathcal{O}_{\mathbb{P}} & \longrightarrow & \mathcal{O}_{\mathbb{P}}(d) \longrightarrow 0 \\
& & \downarrow \varphi & & \downarrow & & \downarrow \\
0 & \longrightarrow & M_d & \longrightarrow & S^d V \otimes \mathcal{O}_{\mathbb{P}} & \longrightarrow & \mathcal{O}_{\mathbb{P}}(d) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & \text{coker } \varphi & \longrightarrow & \mathcal{O}_{\mathbb{P}}^{\oplus c} & \longrightarrow & 0 \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & 0 & & 0 & & 0
\end{array}$$

(Note: Dashed arrows indicate the snake lemma application: from $\ker \varphi$ to $\text{coker } \varphi$ and from M_W to $\mathcal{O}_{\mathbb{P}}^{\oplus c}$.)

We show that the middle short exact sequence are both exact. (The left and right column are clearly exact.)

The middle column is exact since it just tensoring $\otimes \mathcal{O}_{\mathbb{P}}$ with

$$0 \longrightarrow W \longrightarrow S^d V \longrightarrow W/S^d V = \mathbb{C}^{\oplus c} \longrightarrow 0$$

Thus by snake lemma we find that $\ker \varphi = 0$ and $\text{coker } \varphi = \mathcal{O}_{\mathbb{P}}^{\oplus c}$, which give an exact sequence

$$0 \longrightarrow M_W \longrightarrow M_d \longrightarrow \mathcal{O}_{\mathbb{P}}^{\oplus c} \longrightarrow 0$$

Now we form the Eagon-Northgott exact sequence 1.4 of $M_d \twoheadrightarrow \mathcal{O}_{\mathbb{P}}^{\oplus c}$

$$0 \longrightarrow (\wedge^{\binom{n+d}{d}-1} M_d) \otimes (S^{\binom{n+d}{d}-1-c} \mathcal{O}_{\mathbb{P}}^{\oplus c})^* \otimes (\wedge^c \mathcal{O}_{\mathbb{P}}^*) \longrightarrow \dots$$

$$\begin{array}{ccccccc} \dots & \longrightarrow & (\wedge^{c+2} M_d) \otimes (\mathcal{O}_{\mathbb{P}}^*)^{\oplus c} \otimes [\wedge^c (\mathcal{O}_{\mathbb{P}}^*)^{\oplus c}] & \longrightarrow & (\wedge^{c+1} M_d) \otimes [\wedge^c (\mathcal{O}_{\mathbb{P}}^*)^{\oplus c}] & \longrightarrow & M_d \longrightarrow \mathcal{O}_{\mathbb{P}}^{\oplus c} \longrightarrow 0 \\ & & & & \downarrow & \nearrow & \\ & & & & M_W & & \\ & \nearrow & & & \downarrow & & \\ 0 & & & & 0 & & \end{array}$$

Since M_d is 1-regular and free (Freeness from 2.3), we know that $\wedge^k M_d$ is k -regular by 2.5 and thus the term L_p is $(p-2) + (c+1)$ -regular. By 2.2, M_W is $(c+1)$ -regular. In particular $H^1(\mathbb{P}, M_W(c)) = 0$. Finally twist $\mathcal{O}_{\mathbb{P}}(c)$ and take long exact sequence of the below short exact sequence.

$$0 \longrightarrow M_W \longrightarrow W \otimes \mathcal{O}_{\mathbb{P}} \longrightarrow \mathcal{O}_{\mathbb{P}}(d) \longrightarrow 0$$

□

Lemma 2.3. Let A be a (Noetherian) local ring. Assume we have a short exact sequence of A -modules

$$0 \longrightarrow M \longrightarrow N \longrightarrow L \longrightarrow 0$$

such that N, L is free, then M is free.

Proof. N, L is flat by freeness, and thus a simple argument of Tor^A we know that M is flat. In (Noetherian) local ring, flat module is free and we done. □

Example 2.3. (Regularity with respect to a vector bundle) Let X be an irreducible projective variety of dimension n , and $\mathcal{E}$ be a locally free sheaf on X such that for any $x \in X$, there is a section $s \in H^0(X, \mathcal{E})$ such that $x \in Z(s)$ and $\dim Z(s) = 0$, where $Z(s)$ is the zero locus of s i.e. the closed subscheme defined by the ideal sheaf $\mathcal{I}_{Z(s)} = \text{im}(s^\vee : \mathcal{E}^* \rightarrow \mathcal{O}_X)$. Assume that $\mathcal{F}$ is a coherent sheaf on X such that

$$H^i(X, \wedge^i \mathcal{E}^* \otimes \mathcal{F}) = 0$$

for all $i > 0$. Then $\mathcal{F}$ is globally generated.

Proof. We just need to show that

$$H^0(X, \mathcal{F}) \otimes \mathcal{O}_{X,x} \longrightarrow \mathcal{F}_x$$

is surjective for any $x \in X$. Given $x \in X$, by assumption, we choose a section $s \in H^0(X, \mathcal{E})$ such that $s(x) = 0$ and $\dim Z(s) = 0$. Consider the Koszul complex. (after twisting $\mathcal{F}$.)

$$\dots \longrightarrow \wedge^2 \mathcal{E}^* \otimes \mathcal{F} \longrightarrow \wedge^1 \mathcal{E}^* \otimes \mathcal{F} \longrightarrow \mathcal{O}_X \otimes \mathcal{F} \longrightarrow \mathcal{O}_{Z(s)} \otimes \mathcal{F} \longrightarrow 0$$

By the condition $H^i(X, \wedge^i \mathcal{E}^* \otimes \mathcal{F}) = 0$ and the lemma below 2.4, we have

$$H^0(X, \mathcal{F}) \longrightarrow H^0(X, \mathcal{F} \otimes \mathcal{O}_{Z(s)})$$

is surjective. Note that $\dim Z(s) = 0$ and $Z(s)$ is a finite type k -scheme, thus we have

$$Z(s) \cong \coprod_{i=1}^m \text{Spec}(\mathcal{O}_{Z(x), x_i})$$

where $\text{Supp } Z(s) = \{x_1, \dots, x_m\}$ is closed and thus the natural map

$$(\mathcal{F} \otimes \mathcal{O}_{Z(s)})(U) \longrightarrow \prod_{z \in U} (\mathcal{F} \otimes \mathcal{O}_{Z(s)})_z$$

is isomorphism. Consider

$$H^0(X, \mathcal{F} \otimes \mathcal{O}_{Z(s)}) = \bigoplus_{i=1}^m (\mathcal{F} \otimes \mathcal{O}_{Z(s)})_{x_i} = \bigoplus_{i=1}^m (\mathcal{F}_{x_i} \otimes_{\mathcal{O}_{X, x_i}} \mathcal{O}_{Z(s), x_i}) \twoheadrightarrow \bigoplus_{i=1}^m (\mathcal{F}_{x_i} \otimes_{\mathcal{O}_{X, x_i}} k(x))$$

Therefore one has

$$H^0(X, \mathcal{F}) \twoheadrightarrow \mathcal{F}_x \otimes_{\mathcal{O}_{X, x}} k(x)$$

for any $x \in X$ and thus by Nakayama lemma $\mathcal{F}$ is generated by global section at x . $\square$

Lemma 2.4. Let X be a projective scheme, and

$$0 \longrightarrow \mathcal{F}_l \longrightarrow \dots \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F} \longrightarrow 0$$

be a complex of coherent sheaves. Assume that

$$H^1(X, \mathcal{F}_1) = H^2(X, \mathcal{F}_2) = \dots = H^l(X, \mathcal{F}_l) = 0$$

and the above complex is exact off Z , where Z satisfying $\dim Z = 0$ is a closed subset of X . Then the map

$$H^0(X, \mathcal{F}_0) \longrightarrow H^0(X, \mathcal{F})$$

is surjective.

Proof. Omit. See Lazarsfeld appendix B 1.2. $\square$

Example 2.4. (Apply to Grassmannians) Let $\mathbb{G} = \text{Gr}(k, m)$ be the Grassmannian of k -dimensional quotient of an m -dimensional vector space and denote by $\mathcal{E}$ the tautological rank k quotient bundle on $\mathbb{G}$. Suppose that $\mathcal{F}$ is a coherent sheaf on $\mathbb{G}$ satisfying the condition that for every $i > 0$,

$$H^i(\mathbb{G}, \wedge^{i_1} \mathcal{E}^* \otimes \dots \otimes \wedge^{i_{m-k}} \mathcal{E}^* \otimes \mathcal{F}) = 0$$

whenever $i_1 + \dots + i_{m-k} = i$. Then $\mathcal{F}$ is globally generated.

Proof. We just use 2.3 and $\wedge^i(\bigoplus_{j=1}^m \mathcal{F}_j) = \bigoplus_{j_1 + \dots + j_m = i} (\wedge^{j_1} \mathcal{F}_1 \otimes \dots \otimes \wedge^{j_m} \mathcal{F}_m)$. $\square$

Proposition 2.6. (Criterion for very ampleness) Let X be an irreducible projective variety of dimension n , $\mathcal{N}$ a 0-regular line bundle and $\mathcal{L}$ a free ample line bundle, then $\mathcal{N} \otimes \mathcal{L}$ is very ample.

Proof. Since $\mathcal{N} \otimes \mathcal{L}$ is free, by the lemma below 2.5 we know that it suffice to proof $\mathcal{N} \otimes \mathcal{L} \otimes \mathfrak{m}_x$ is globally generated for every closed $x \in X$. We choose a section $s \in \mathcal{E} := \mathcal{L}^{\oplus n}$ such that $Z(s) \ni x$ is a zero-dimensional closed subscheme and $\mathcal{L} \otimes \mathcal{N} \otimes \mathcal{I}_{Z(s)}$ is 0-regular. Consider the exact sequence

$$0 \longrightarrow \mathcal{N} \otimes \mathcal{I}_{Z(s)}(k) \longrightarrow \mathcal{N} \otimes \mathfrak{m}_x(k) \longrightarrow [\mathcal{N} \otimes \mathfrak{m}_x(k) / \mathcal{I}_{Z(s)}](k) \longrightarrow 0$$

Note that $[\text{Supp } \mathcal{N} \otimes \mathfrak{m}_x(k) / \mathcal{I}_{Z(s)}](k) \subseteq Z(s)$ and thus $\dim \text{Supp } \mathcal{N} \otimes \mathcal{I}_{Z(s)} / \mathfrak{m}_x(k) = 0$. By Grothendieck vanishing theorem

$$H^i(X, [\mathcal{N} \otimes \mathfrak{m}_x(k) / \mathcal{I}_{Z(s)}](k)) = 0$$

for all $i > 0$ and $k \in \mathbb{Z}$. Chasing through the exact sequence one can deduce that

$$H^i(X, \mathcal{N} \otimes \mathcal{I}_{Z(s)}(k)) = H^i(X, \mathcal{N} \otimes \mathfrak{m}_x(k))$$

for all $i > 1, k \in \mathbb{Z}$ and

$$H^1(X, \mathcal{N} \otimes \mathcal{I}_{Z(s)}(k)) \longrightarrow H^1(X, \mathcal{N} \otimes \mathfrak{m}_x(k)) \longrightarrow 0$$

From the 0-regularity of $\mathcal{N} \otimes \mathcal{L} \otimes \mathcal{I}_{Z(s)}$ we know that $\mathcal{N} \otimes \mathcal{L} \otimes \mathfrak{m}_x$ is 0-regular. Thus $\mathcal{L} \otimes \mathcal{N} \otimes \mathcal{I}_{Z(s)}$ is generated by global section by Mumford's theorem. Now we proceed with two steps to ensure the condition we mentioned above.

1. Choose section $s \in \mathcal{E}$ such that $\dim Z(s) = 0$. We consider that finite morphism defined by $\mathcal{L}$.

$$\phi_{|\mathcal{L}|} : X \longrightarrow \mathbb{P}H^0(X, \mathcal{L}) =: Y$$

Then we have that the section of $H^0(X, \mathcal{L})$ that vanishing at x corresponding to all hyperplanes in Y containing $\phi_{|\mathcal{L}|}(x)$, or more formally,

$$H^0(X, \mathcal{L} \otimes \mathfrak{m}_x) = H^0(X, \phi_{|\mathcal{L}|}^* \mathcal{O}_Y(1) \otimes \mathfrak{m}_x) = H^0(Y, \phi_*(\phi_{|\mathcal{L}|}^* \mathcal{O}_Y(1) \otimes \mathfrak{m}_x)) = H^0(Y, \mathcal{O}_Y(1) \otimes \mathfrak{m}_{\phi_{|\mathcal{L}|}(x)}) \neq 0.$$

Now choose any nonzero section $s_1 \in H^0(X, \mathcal{L} \otimes \mathfrak{m}_x)$, we know that $Z(s_1) \subsetneq X$ and thus we get a lower dimension subscheme. Now we consider $Z(s_1) = \cup Z_i$ be the irreducible decomposition. Note that $\phi_{|\mathcal{L}|}$ is finite by Lazarsfeld corollary 1.2.15. and thus $\dim \phi_{|\mathcal{L}|}(Z_i) \geq 1$. If we denote the generic point of Z_i by $\mathfrak{p}_i$, then $\mathfrak{p}_i \neq \phi_{|\mathcal{L}|}(x)$. Now by prime avoidance lemma for $\mathfrak{p}_i$, $I = H^0(X, \mathcal{L} \otimes \mathfrak{m}_x)$ and ring $A = \Gamma_*(Y)$. we know that there is a section $s_2 \in H^0(X, \mathcal{L} \otimes \mathfrak{m}_x)$ such that $s_2 \notin \cup \mathfrak{p}_i$ i.e. s_2 do not passing through the generic point of Z_i and thus we get a lower dimension. And continue this process we get desired section s .

2. $\mathcal{L} \otimes \mathcal{N} \otimes \mathcal{I}_{Z(s)}$ is 0-regular. We consider the Koszul complex of s . (After tensoring with $\mathcal{N} \otimes \mathcal{L}^{\otimes(1-i)}$.)

$$\cdots \longrightarrow \mathcal{N}(-i-2)^{\binom{n}{3}} \longrightarrow \mathcal{N}(-i-1)^{\binom{n}{2}} \longrightarrow \mathcal{N}(-i)^{\binom{n}{1}} \longrightarrow \mathcal{N} \otimes \mathcal{I}_{Z(s)}(1-i) \longrightarrow 0$$

Since $\mathcal{N}$ is 0-regular, we know that

$$H^i(X, \mathcal{N}(-i)) = H^{i+1}(X, \mathcal{N}(-1-i)) = \cdots = 0$$

By the lemma below 2.6, we know that $H^i(X, \mathcal{N} \otimes \mathcal{I}_{Z(s)}(1-i)) = 0$, which just says that $\mathcal{N} \otimes \mathcal{L} \otimes \mathcal{I}_{Z(s)}$ is zero regular.

□

Lemma 2.5. Let X be a scheme, and $\mathcal{L}$ a free invertible sheaf on X . If $\mathcal{L} \otimes \mathfrak{m}_x$ is generated by global section for any closed point $x \in X$. Then $\phi_{|\mathcal{L}|} : X \rightarrow \mathbb{P}H^0(X, \mathcal{L})$ is a closed embedding. Here $\mathfrak{m}_x$ is the ideal sheaf corresponding to $\{x\}$.

Proof. We use Hartshorne chapter 2 proposition 7.3. We just need to proof that $|\mathcal{L}|$ separates points and tangent vector.

1. For any two distinct closed points $x \neq y \in X$, we consider the stalk of $\mathcal{L} \otimes \mathfrak{m}_x$ at y . Note that $(\mathfrak{m}_x)_y = \mathcal{O}_{X,y}$ since $\text{Supp } \mathcal{O}_X / \mathfrak{m}_x = \{x\}$. Therefore we have

$$(\mathcal{L} \otimes \mathfrak{m}_x)_y = \mathcal{L}_y \otimes_{\mathcal{O}_{X,y}} \mathcal{O}_{X,y} = \mathcal{L}_y$$

the globally generated of $\mathcal{L} \otimes \mathfrak{m}_x$ implies that there is a section

$$s \in H^0(X, \mathcal{L} \otimes \mathfrak{m}_x) \subseteq H^0(X, \mathcal{L} \otimes \mathcal{O}_X) = H^0(X, \mathcal{L})$$

(The above inclusion is given by applying $\otimes \mathcal{L}$ and $H^0(X, -)$ to $0 \rightarrow \mathfrak{m}_x \rightarrow \mathcal{O}_X \rightarrow k(x) \rightarrow 0$) such that $s_y \notin \mathfrak{m}_{X,x} \cdot \mathcal{L}_y$. From $s \in H^0(X, \mathcal{L} \otimes \mathfrak{m}_x)$ we know that $s_x \in \mathfrak{m}_{X,x} \cdot \mathcal{L}_x$.

2. We proof that $\frac{\mathfrak{m}_{X,x} \cdot \mathcal{L}_x}{\mathfrak{m}_{X,x}^2 \cdot \mathcal{L}_x}$ is generated by $\{s \in H^0(X, \mathcal{L}) \mid s_x \in \mathfrak{m}_{X,x} \cdot \mathcal{L}_x\}$. This time we consider

$$(\mathcal{L} \otimes \mathfrak{m}_x)_x \otimes_{\mathcal{O}_{X,x}} k(x) = \mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} \mathfrak{m}_{X,x} \otimes k(x) = \mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} \frac{\mathfrak{m}_{X,x}}{\mathfrak{m}_{X,x}^2} = \frac{\mathfrak{m}_{X,x} \cdot \mathcal{L}_x}{\mathfrak{m}_{X,x}^2 \cdot \mathcal{L}_x}$$

the globally generated property just says the surjectivity of the following map

$$\{s \in H^0(X, \mathcal{L}) \mid s_x \in \mathfrak{m}_{X,x} \cdot \mathcal{L}_x\} = H^0(X, \mathcal{L} \otimes \mathfrak{m}_x) \twoheadrightarrow (\mathcal{L} \otimes \mathfrak{m}_x)_x \otimes_{\mathcal{O}_{X,x}} k(x) = \frac{\mathfrak{m}_{X,x} \cdot \mathcal{L}_x}{\mathfrak{m}_{X,x}^2 \cdot \mathcal{L}_x}$$

□

Lemma 2.6. Let X be a projective scheme, and

$$0 \longrightarrow \mathcal{F}_l \longrightarrow \cdots \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F} \longrightarrow 0$$

be a complex of coherent sheaves. Assume that

$$H^i(X, \mathcal{F}_0) = H^{i+1}(X, \mathcal{F}_1) = \cdots = H^{i+l}(X, \mathcal{F}_l) = 0$$

and the above complex is exact off Z , where Z satisfying $\dim Z \leq i$ is a closed subset of X . Then

$$H^i(X, \mathcal{F}) = 0$$

Proof. Omit. See Lazarsfeld appendix B 1.2. □

Example 2.5. (Adjoint bundles) Let X be a smooth projective variety of dimension n , D be a free ample divisor and P be any nef divisor on X , then we have that the line bundle

$$\mathcal{L}_k = \mathcal{O}_X(K_X + kD + P)$$

is

1. free if $k \geq n + 1$;
2. very if $k \geq n + 2$.

Proof.

1. We use the Kodaria's vanishing theorem. For $0 < i < n + 1$, we have

$$H^i(X, \mathcal{O}_X(K_X + kD + P - iD)) = H^i(X, \mathcal{O}_X(K_X + (k - i)D + P)) = 0$$

by applying 2.3 to the ample divisor $(k - i)D + P$ since ample + nef = ample. For $i \geq n + 1$, we use Grothendieck's vanishing theorem. In particular, this implies that $\mathcal{L}_k$ is 0-regular when $k \geq n + 1$. Thus by Mumford's theorem 2.1 we know that $\mathcal{L}_k$ is free.

2. For part 2, we use the criterion describe as above 2.6 for

$$\mathcal{L}_k = \mathcal{L}_{k-1} \otimes \mathcal{O}_X(D)$$

□

Theorem 2.3. (Kodaria's vanishing theorem) Let X be a smooth projective variety of dimension n and D be an ample divisor, then we have

$$H^i(X, \mathcal{O}_X(K_X + D)) = 0$$

for all $i > 0$.

Proof. See Lazarsfeld theorem 4.2.1. □

3 Regularity and Complexity

Definition 9. Let $\mathbb{P} = \text{Proj}.k[x_0, \dots, x_r]$ be the projective space of dimension r , $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}$ and

$$S = k[x_0, \dots, x_r], \quad F = \Gamma_*(\mathcal{F}) = \bigoplus_{k=0}^{\infty} H^0(\mathbb{P}, \mathcal{F}(k)).$$

Assume more that F is bounded below. By 1.2 F admits a minimal graded free resolution with no nonzero constant part in the maps.

$$0 \longrightarrow E_{r+1} \longrightarrow \dots \longrightarrow E_2 \longrightarrow E_1 \longrightarrow E_0 \longrightarrow F \longrightarrow 0$$

If we denote

$$E_p = \bigoplus_i S(-a_{p,i})$$

then $a_{p,i}$ uniquely determined by F 1.3. We define

$$a_p = a_p(\mathcal{F}) = \max_i \{a_{p,i}\}$$

i.e. the maximal degree of generators of p -th module of syzygy of F .

Definition 10. For a subscheme X of $\mathbb{P}$, we define that X is m -regular if the ideal sheaf $\mathcal{I}_X$ of X is m -regular. And we define the regularity of X to be

$$\text{reg}(X) := \text{reg}(\mathcal{I}_X).$$

Theorem 3.1. The coherent sheaf $\mathcal{F}$ on $\mathbb{P}$ is m -regular iff

$$a_p(\mathcal{F}) \leq p + m$$

for all p

Proof. If $a_{p,i} \leq p + m$ for all i and p , then by 2.2 we know that $\mathcal{F}$ is m -regular since $S(-a_{p,i})^\sim = \mathcal{O}_{\mathbb{P}}(-a_{p,i})$ is $m + p$ -regular. Conversely, if $\mathcal{F}$ is m -regular, then by Mumford's theorem we know that

$$H^0(\mathbb{P}, \mathcal{F}(m)) \otimes H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(k)) \twoheadrightarrow H^0(\mathbb{P}, \mathcal{F}(m+k))$$

is surjective. Because $H^0(\mathbb{P}, \mathcal{F}(j))$ are all finitely generated k -module, we can choose the generators of degree $\leq m$ to generate F , which gives an exact sequence

$$0 \longrightarrow F_1 \longrightarrow \bigoplus_i S(-a_{0,i}) \longrightarrow F \longrightarrow 0$$

By choice we have $a_{0,i} \leq m$. Now just proof that $F_1^\sim$ is $m+1$ -regular, and then we can continue this process. This is just by taking $\sim$, twist $\mathcal{O}_{\mathbb{P}}(m+1-i)$ and taking cohomology in above short exact sequence

$$0 = H^{i-1}(\mathbb{P}, \mathcal{F}(m - (i-1))) \longrightarrow H^i(\mathbb{P}, F_1^\sim(m+1-i)) \longrightarrow \bigoplus_j H^{i-1}(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m - (i-1) + a_{0,j})) = 0$$

□

Example 3.1. Let $X \subseteq \mathbb{P}$ be the complete intersection of hypersurfaces of degree $d_1, \dots, d_e$ and $\mathcal{I}_X$ be the ideal sheaf of X . Then we have

$$\text{reg}(X) = 1 - e + \sum_{i=1}^e d_i$$

Proof. Let $X = \bigcap_{i=1}^e V_+(f_i) = V(f_1, \dots, f_e)$ be the complete intersection with $\deg f_i = d_i$, then we can consider the exact Koszul complex of $(f_1, \dots, f_e) \in H^0(\mathbb{P}, \mathcal{E})$, where $\mathcal{E} = \bigoplus_{i=1}^e \mathcal{O}_{\mathbb{P}}(d_i)$.

$$0 \longrightarrow \wedge^e \mathcal{E}^* \longrightarrow \dots \longrightarrow \wedge^2 \mathcal{E}^* \longrightarrow \wedge^1 \mathcal{E}^* \longrightarrow \mathcal{I}_X \longrightarrow 0$$

WLOG, we assume that $d_e \geq \dots \geq d_1 \geq 1$, then we know that $\wedge^k \mathcal{E}^*$ is $d_e + \dots + d_{e-k+1}$ -regular. Thus by 2.2 we know that $\mathcal{I}_X$ is $(d_e + \dots + d_1 - e + 1)$ -regular. Finally by above theorem for $p = e - 1$ we know that

$$\text{reg}(X) = d_e + \dots + d_1 - e + 1$$

□

Example 3.2. If $i : X \hookrightarrow \mathbb{P}$ is a subvariety of dimension n , then for $m > 0$, X is m -regular iff

$$H^i(\mathbb{P}, \mathcal{I}_X(m-i)) = 0$$

for all $1 \leq i \leq n+1$.

Proof. Consider the short exact sequence

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_{\mathbb{P}} \longrightarrow i_* \mathcal{O}_X \longrightarrow 0$$

Twisting $\mathcal{O}_{\mathbb{P}}(m-i)$ and taking cohomology we get

$$0 = H^{i-1}(\mathbb{P}, \mathcal{O}_{\mathbb{P}}((m-1) - (i-1))) \longrightarrow H^{i-1}(X, \mathcal{O}_X(m-i)) = 0$$

$$\xrightarrow{\delta} H^i(\mathbb{P}, \mathcal{I}_X(m-i)) \longrightarrow H^i(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m-i)) = 0$$

These terms are 0 by the 0-regularity of $\mathcal{O}_{\mathbb{P}}$, $m > 0$ and Grothendieck vanishing theorem for X when $i-1 > n$. □

Example 3.3. If $i : X \hookrightarrow \mathbb{P}$ is a subvariety of dimension n , then

$$\text{reg}(X) \geq 0$$

with $\text{reg}(X) = 0$ iff $X = \emptyset$.

Proof. If $X = \emptyset$ then $\mathcal{I}_X = \mathcal{O}_{\mathbb{P}}$, which is 0-regular. For $X \neq \emptyset$, then $\mathcal{I}_X \neq \mathcal{O}_{\mathbb{P}}$ and we consider its corresponding saturated ideal $I_X = \Gamma_*(\mathcal{I}_X)$ of $S = \Gamma_*(\mathcal{O}_{\mathbb{P}})$. We choose a minimal graded free resolution of I_X , then we know that

$$\text{reg}(\mathcal{I}_X) = \max_p \{a_p(\mathcal{I}_X) - p\} \geq a_0(\mathcal{I}_X) \geq 1$$

since $I_X \neq S$ and thus the generator of I_X must be polynomial of degree ≥ 1 . $\square$

Example 3.4. (Regularity of finite sets) Let $X \subseteq \mathbb{P}$ be a finite subset of d distinct closed points $X = \{x_1, \dots, x_d\}$. Then X is d -regular, and if x_i are collinear, then X is not $(d-1)$ -regular. So in this case

$$\text{reg}(X) = d$$

Proof. Consider the short exact sequence

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_{\mathbb{P}} \longrightarrow \bigoplus_{i=1}^d k(x_i) \longrightarrow 0$$

For $i > 1$, this is trivial by twisting $\mathcal{O}_{\mathbb{P}}(d-i)$ and taking cohomology since $\mathcal{O}_{\mathbb{P}}$ is 0-regular and

$$H^k(\mathbb{P}, \bigoplus_{i=1}^d k(x_i)) = 0$$

by Grothendieck vanishing theorem for all $k > 1$. So we focus on the case $i = 1$. By twisting $\mathcal{O}_{\mathbb{P}}(d-1)$ and taking cohomology we just need to proof that

$$\begin{aligned} H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(d-1)) &\longrightarrow H^0(\mathbb{P}, \bigoplus_{i=1}^d k(x_i)) \cong k^{\oplus d} \\ s &\longmapsto (s(x_i)) \end{aligned}$$

is surjective. But the surjectivity follows from that any distinct d points determine a polynomial of degree $\leq d-1$.

If x_i are collinear, then we consider the line $\mathbb{P}^1 \cong L \supseteq X$ in $\mathbb{P}$. Now assume that X is $d-1$ -regular, then we know that

$$H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(d-1)) \twoheadrightarrow H^0(\mathbb{P}, \bigoplus_{i=1}^d k(x_i)) \cong k^{\oplus d}$$

is surjective. But apply $H^0(\mathbb{P}, -)$ to $\mathcal{O}_{\mathbb{P}} \rightarrow \mathcal{O}_L \rightarrow \mathcal{O}_X$ we know that this map can be decomposed as

$$H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(d-1)) \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(d-2)) \twoheadrightarrow H^0(\mathbb{P}, \bigoplus_{i=1}^d k(x_i)) \cong k^{\oplus d}$$

Now

$$\dim H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(d-2)) = d-1, \quad \dim k^{\oplus d} = d$$

contradicts to the surjectivity. $\square$

Example 3.5. (Regularity of a disjoint union) Let $X, Y \subseteq \mathbb{P}$ be a disjoint subschemes such that $X \cap Y = \emptyset$. If X is m -regular and Y is l -regular, then $X \amalg Y$ is $(m+l)$ -regular.

Proof. We consider the short exact sequence (when $X \cap Y = \emptyset$, $\mathcal{I}_X + \mathcal{I}_Y = \mathcal{O}_{\mathbb{P}}$ since $\text{Supp } \mathcal{O}_{\mathbb{P}}/(\mathcal{I}_X + \mathcal{I}_Y) = \emptyset$.)

$$0 \longrightarrow \mathcal{I}_{X \cup Y} \longrightarrow \mathcal{I}_X \oplus \mathcal{I}_Y \longrightarrow \mathcal{I}_X + \mathcal{I}_Y = \mathcal{O}_{\mathbb{P}} \longrightarrow 0$$

Thus $\mathcal{I}_{X \cup Y} = \mathcal{I}_X \cap \mathcal{I}_Y = \mathcal{I}_X \cdot \mathcal{I}_Y = \mathcal{I}_X \otimes \mathcal{I}_Y$ when $X \cap Y = \emptyset$ (The last two equality use $X \cap Y = \emptyset$). And the conclusion follows from 2.4. (Note that the proposition holds when for any $x \in X$ either $(\mathcal{I}_X)_x$ or $(\mathcal{I}_Y)_x$ is locally free, since in the proof we only need the exactness after tensoring the linear resolution.) $\square$